MST++ Thursday+STream+with+ Answers

Derivative Securities (University of Melbourne)



Scan to open on Studocu

Studocu is not sponsored or endorsed by any college or university
Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

FNCE30007 – Derivative Securities
MID-SEMESTER TEST – SEMESTER 2, 2023 (THURSDAY) – WITH ANSWERS

Question 1 (1 mark)
Henry Winkler sells a futures contract on an asset when the futures price is $1,500. Each contract is on 100 units of the asset. The contract is closed out when the futures price is $1,480. Which of the following is true

A. Henry has made a gain of $4,000
B. Henry has made a loss of $4,000
C. Henry has made a gain of $2,000
D. henry has made a loss of $2,000
E. One cannot tell unless we know the spot price of the asset when the futures position is closed out

An investor who sells (has a short position) makes gains when a futures price decreases.

Question 2 (1 mark)
It is now 15th June 2020, you are bullish on the gold price for the next 6 months and willing to speculate by using $197,000 to take the appropriate position in 6-month futures contract. The current spot price and December futures price of gold are $1,950 and $1,975 per ounce, respectively. Each contract is on 100 ounces of gold. The required initial margin is $9,850 per contract. The maintenance margin per contract is $7,350. At the end of the first day of trading, the futures contract settles at $1,990 per ounce. What is the balance of your margin account at the end of that day?

A. $167,000
B. $227,000
C. $195,500
D. $197,000
E. None of the other answers

With $197,000, you are able to long 197,000/9850 = 20 contracts. The price has increased by $15. Because you have a long position you gain 15×100x20 or $30,000. The balance in the margin account therefore goes up from $197,000 to $227,000.

Question 3 (1 mark)
Bloomberg Commodity News reports that the level of copper inventories has recorded a very large and unexpected drop. All other things equal, what impact will such news have on prices?

A. The 6-month copper futures price as a percentage of the spot price increases
B. The copper spot price as a percentage of the 6-month futures price increases
C. The copper spot price as a percentage of the 6-month futures price declines
D. The 6-month copper futures price as a percentage of the spot price stays the same
E. The copper futures curve becomes necessarily inverted due to the trading by arbitrageurs

When inventories fill up, the convenience yield drops and the spot price as a percentage of the futures price declines. The opposite occurs when inventory drops significantly and unexpectedly.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Question 4 (1 mark)
Which of the following scenarios increases basis risk?

A. A large difference between the futures prices when the hedge is put in place and when it is closed out
B. Differences between the asset underlying the futures contract and the asset whose price generates the hedger's exposure
C. An expected decrease of the futures price relatively to the spot price
D. A small difference between the futures prices when the hedge is put in place and when it is closed out
E. None of the other scenarios increase basis risk

Basis is the difference between futures and spot at the time the hedge is closed out. This increases as the time between the date when the futures contract is put in place and the delivery month increases. (C is not therefore correct). It also increases as the asset underlying the futures contract becomes more different from the asset being hedged. (B is therefore correct.)

Question 5 (2 marks)

The Susan Ross Foundation has invested its endowment money in a large diversified portfolio of US equities. The portfolio, currently worth $30,000,000, is fairly aggressive as it currently displays a beta of 1.35 with respect to the S&P500 spot index. As the newly hired assistant to the portfolio manager, you are told that the Foundation wants to lower the portfolio risk by reducing beta to 0.7 over the next 8 months. Armed with your knowledge in derivatives, you consider using S&P500 futures contacts offered by CME (whose contract size is $250 times the index). You also assess that the dividend yield on the S&P500 index is 2% per year and the current risk-free rate is 0.4% per year (both dividend yield and interest rates are expressed with continuous compounding). At the end of today the S&P500 spot closed at 1,350. Assume that today is July 16th and S&P500 Index futures are available for March, June, September and December of each year

Which of the following is the most appropriate hedging strategy needed to achieve the Foundation's target beta?

a. Short 90 March contracts
b. Long 58 March contracts
c. Short 90 December contracts
d. Short 58 March contracts
e. Long 58 December contracts and Short 58 March contracts

$S_t = 1350$ r=0.4% q=2% T=8/12 for March contract
$F_t = S_t*\exp[(r-q)*T] = 1350*\exp[(0.4\%-2\%)*8/12] = \$1335.68$
$ Value of one Futures Contract is = $250*1335.68 = $333,920

Fund should short (1.35 – 0.7)*30,000,000 / 333,920 = 58.38 i.e., 58 S&P500 futures contracts

Question 6 (2 marks)
On July 5, a stock index futures contract with expiration on December 20 settles at 390.84. On the same day, the spot index closes at 392.54. Assume that each index point is worth $1. The risk-free rate of interest is 1.5% per annum and the dividend yield on the spot index is 3.5% per annum (both rates are continuously compounded). Transaction costs are negligible. Then:

a. Go long the futures, short the spot, invest the proceeds and realize an arbitrage profit of $1.88
b. Go short the futures, borrow to long the spot and realize an arbitrage profit of $1.70
c. The futures and the spot are fairly priced and, hence, there is no arbitrage opportunity
d. Go short the futures, borrow to long the spot and realize an arbitrage profit of $1.88

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

e. One cannot tell if there is an arbitrage opportunity or not, as we don't know what the spot price of the index will be at the expiration of the futures contract

Answer:
S=392.54, r=1.5%, q=3.5%, T=5.5/12
No-arbitrage futures price = 392.54e^(1.5%-3.5%)*(5.5/12) = 388.9582
Since the market price is higher than the no-arbitrage price, hence it is over-priced. One should short futures, long spot with borrowed funds, the arbitrage profits realized will be 390.84-388.9582=$1.88

Question 7 (1 mark)
Which of the following statements is correct?

A. The optimal hedge ratio is chosen to minimize the variance of the hedged position
B. The optimal hedge ratio is always less than 1 if the hedge is not perfect
C. The optimal hedge ratio is always equal to 1 if changes in the spot price are perfectly positively correlated with changes in the futures price
D. When using index futures to hedge an equity portfolio, the optimal hedge ratio is always lower than the beta of the portfolio
E. None of the other statements is correct

Question 8 (1 mark)
Johnny Walker wants to hedge his currency exposure by using the forward FX market. In particular, being based in Wellington, New Zealand, he is concerned about fluctuation of the New Zealand dollar (NZD) value vs. the US dollar (USD) over the next six months. Johnny observes that the risk-free rate is 7% in New Zealand and 5% in the US (both interest rates are quoted in continuously compounded terms per annum). Johnny looks for quotes on the forward market but only finds a quote of 1.5202 NZD per USD for a 3-month maturity. He, then, calls up his bank to ask for a 6-momth forward rate quote.

Ignoring transactions costs, which one of the following 6-month forward quotes does NOT represent an arbitrage opportunity?

A. 1.5126 NZD per USD
B. 0.6511 USD per NZD
C. 1.6513 NZD per USD
D. 0.6645 USD per NZD
E. 1.5278 NZD per USD

DC = NZD, FC = USD, r = 7%, rf = 5%, F0=S0*exp[(r-rf)*T]
For 3-months contract:1.5202 = S0*exp[(0.07-0.05)*0.25] => S0=1.512618
For 6-months contract: F0 = 1.512618*exp[(0.07-0.05)*0.5] = 1.5278

Question 9 (1 mark)
We watched the trading scene from the movie Trading Places. Which of the following statements is correct:

a) After the announcement by the Department of Agriculture, the vast majority of traders wanted to go long FCOJ (Frozen Concentrated Orange Juice) contracts to get out of the large short positions they had accumulated.
b) Before the start of the trading session, Winthorpe and Valentine had purchased FCOJ which they, then, sold through the futures market.
c) The Dukes faced a margin call as they were net short at the end of the trading session.
d) Winthorpe and Valentine initially opened short futures positions and, after the futures price went down, they closed out by going long.
e) The Duke brothers thought the orange crop was going to be strong and, therefore, wanted to build a large long position in FCOJ futures.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Question 10 (1 mark)
During the lectures we covered the FT articles "Shale oil drillers left exposed after pulling back price hedges" and "Gold bulls deeply suspicious of hedging". Which of the following statements is correct?

A. Rather than using futures for hedging, many gold producers preferred (at the time the "Gold Bulls" article was written) to take long and, thus, speculative futures positions.
B. At the time the "Gold Bulls" article was written, most analysts were recommending oil producers to take long position in gold futures.
C. In 2022 US oil producers earned profits on their hedges.
D. In 2023 many US oil producers decided to reduce their hedging of crude oil price risk as they expected the spot price of crude to drop.
E. Even after a drop in oil prices during the first few months of 2023, many producers are still choosing not to hedge.

Question 11 (1 mark)
Which of the following statements is correct?

a) Futures contracts are preferred over forward contracts by hedgers because gains are realized daily.
b) A speculator expecting a drop in corn inventory to very low levels over the coming year should short corn futures.
c) Futures on stock indices are always cash settled.
d) Hedgers always prefer forward contracts to futures contracts as the former are tailored to the hedgers' needs.
e) None of the other statements is correct.

Question 12 (1 mark)
Which of the following statements is NOT true?

A. Gold and silver have always positive cost of carry
B. Investment assets are held by significant numbers of investors for investment purposes
C. All agricultural commodities are consumption assets
D. Platinum pays no income: therefore its cost of carry is always negative
E. The no-arbitrage forward price of an investment asset can be obtained from the spot price, interest rates, the income paid on the spot asset and the maturity of the contract.

Question 13 (1 mark)
On March 1 the spot price of a commodity is $1,000 and the December futures price is $1,020. On November 1 the spot price is $980 and the December futures price is $981. A producer, wanting to hedge 70% of its underlying exposure, entered into December futures contracts on March 1 to hedge the sale of the commodity, which took place on November 1. The producer, then, closed out its futures position on November 1. What is the producer's total gain/loss *per unit* from the hedged position?

A. $19
B. - $39
C. $20
D. $7.3
E. None of the other answers is correct

The producer of the commodity takes a short futures position.
The gain/loss from spot position: 980-1000 = -20
The gain/loss from futures position: 1020-981 = 39, with hedge ratio of 0.7, the gain/loss = 27.3
The total gain/loss from hedge: -20+27.3 = 7.3.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)